

SUKHINICHI, Russian Federation

WMO: 277070

Lat: 54.100N

Lon: 35.350E

Elev: 781

StdP: 14.29

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.2	-4.7	-14.6	2.6	-9.4	-9.2	3.4	-4.2	19.0	28.2	17.1	24.6	3.1	270	0.529

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.7	84.0	68.5	80.6	67.0	77.5	65.7	70.1	80.7	68.5	78.1	67.0	75.6	4.4	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.5	100.5	75.0	65.1	95.5	73.5	63.5	90.4	71.9	34.5	80.8	33.2	78.3	31.9	75.7	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.6	14.0	12.0	-15.9	88.3	6.0	3.8	-20.3	91.1	-23.8	93.3	-27.2	95.4	-31.6	98.2		
(5)			-16.3	72.6	6.0	1.8	-20.6	73.9	-24.1	75.0	-27.4	76.0	-31.7	77.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.5	19.3	19.5	29.0	44.6	55.1	61.7	66.5	62.9	53.2	42.4	30.9	22.7		
	DBStd	19.00	11.15	12.03	8.36	8.71	8.49	6.59	5.95	6.48	7.00	7.57	9.29	11.39		
	HDD50	4428	950	853	652	208	46	2	0	0	45	255	572	845		
	HDD65	8436	1415	1273	1117	612	321	140	53	116	358	699	1022	1310		
	CDD50	1679	0	0	0	46	204	354	513	401	141	20	0	0		
	CDD65	213	0	0	0	0	14	42	101	52	4	0	0	0		
	CDH74	1596	0	0	0	6	110	277	741	434	28	0	0	0		
	CDH80	384	0	0	0	1	12	43	193	134	1	0	0	0		
(14) Wind	WSAvg	4.9	5.7	5.7	5.7	4.9	4.5	4.2	3.7	4.1	4.0	5.1	5.5	5.8		
(15) Precipitation	PrecAvg	25.4	1.7	1.4	1.4	1.5	2.5	2.8	3.1	2.5	2.3	2.4	1.9	1.7		
	PrecMax	34.2	3.1	2.8	3.0	2.7	5.0	5.0	6.1	5.9	6.1	4.8	3.4	4.2		
	PrecMin	17.8	0.4	0.5	0.4	0.2	0.7	1.2	0.9	0.2	0.3	0.4	0.0	0.6		
	PrecStd	3.6	0.6	0.6	0.7	0.7	1.1	1.1	1.6	1.5	1.1	1.3	0.8	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.5	39.0	53.8	73.2	81.7	84.3	90.2	91.1	77.7	67.1	52.3	45.4		
		MCWB	37.8	35.8	44.2	55.5	65.2	68.7	69.9	67.6	63.0	55.4	49.0	44.0		
	2%	DB	36.0	36.7	47.5	67.4	77.2	80.6	84.8	83.1	72.6	60.7	47.4	41.4		
		MCWB	34.5	34.6	40.3	52.9	63.1	67.0	68.7	68.6	60.9	55.3	45.3	39.9		
	5%	DB	34.0	35.1	42.9	62.8	73.3	77.1	81.3	78.3	67.9	56.3	44.9	37.3		
		MCWB	32.9	33.5	37.6	50.9	60.3	65.2	68.0	66.1	58.7	52.0	43.3	36.4		
	10%	DB	32.4	33.8	39.5	58.6	69.6	73.7	78.0	74.4	64.2	53.0	42.2	34.8		
		MCWB	31.5	32.7	35.8	48.7	58.0	63.9	67.0	63.9	57.1	49.7	41.1	33.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.5	37.4	45.1	56.9	67.4	70.8	72.4	71.6	64.8	58.8	50.0	44.3		
		MCDB	39.4	38.5	53.0	69.8	79.3	81.3	83.4	82.3	74.6	63.6	51.8	45.1		
	2%	WB	34.8	35.2	41.2	54.2	64.2	68.6	70.5	69.3	62.1	55.7	45.9	40.1		
		MCDB	35.6	36.0	46.2	64.2	74.5	77.9	81.4	79.9	69.6	59.8	46.8	41.2		
	5%	WB	33.0	33.9	38.5	52.1	61.7	66.7	69.1	67.3	59.8	52.7	43.6	36.5		
		MCDB	33.9	34.7	42.0	60.4	70.9	75.1	78.8	76.8	65.9	56.0	44.7	37.3		
	10%	WB	31.7	32.7	36.3	49.8	59.5	64.7	67.6	65.1	57.7	50.1	41.0	34.0		
		MCDB	32.4	33.7	39.0	57.2	67.9	72.3	76.1	72.9	63.1	52.7	42.0	34.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	10.6	12.1	15.7	17.3	16.2	16.7	17.0	15.1	11.4	7.4	7.6		
		MCDBR	6.4	7.6	15.9	21.6	21.9	20.4	21.0	22.0	20.2	16.4	10.1	7.6		
	5% WB	MCWBR	6.4	6.8	10.4	10.8	10.8	10.2	8.9	9.9	10.4	10.8	8.6	7.1		
		MCWBR	6.1	7.3	13.8	19.2	19.5	18.6	18.8	20.1	17.9	14.7	8.9	7.2		
(40) Clear-Sky Solar Irradiance	taub		0.288	0.324	0.346	0.414	0.390	0.378	0.413	0.424	0.382	0.351	0.326	0.291		
	taud		2.129	2.062	2.117	2.134	2.308	2.362	2.286	2.243	2.330	2.371	2.247	2.143		
	Ebn at Noon		203	234	259	256	272	276	263	251	247	228	189	174		
	Edh at Noon		21	32	38	43	38	36	39	38	31	24	19	17		
(44) All-Sky Solar Radiation	RadAvg		225	523	979	1274	1712	1798	1725	1438	945	456	182	134		
	RadStd		48	71	130	99	121	153	156	161	106	83	36	36		

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+1.15	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+196	N/A	(46)
(47)	Regional (7 neighbors)	+1.19	N/A	N/A	N/A	N/A	N/A	N/A	-355	+167	N/A	(47)

Nomenclature: See separate page